COSMOLOGY

Problem Set 4: Inflation

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These are the end of chapter problems for Chapter 4 of the notes. Equation numbers refer to the full notes.

1. Normalizing The Inflaton Mass

For $V = \frac{1}{2}m^2\phi^2$, use $\Delta^2_{\mathcal{R}} = 2.1\times10^{-9}$ at $N_* = 60$ to show that $m \approx 6\times10^{-6}M_{\rm Pl} \approx 1.4\times10^{13}$ GeV. What is the Hubble rate during inflation, and what is the energy density $V^{1/4}$ when the CMB scales exit the horizon?

2. The Starobinsky Model

The Starobinsky potential is $V = V_0\left(1 - e^{-\sqrt{2/3}\phi/M_{\rm Pl}}\right)^2$.

a). Show that for large ϕ, $\epsilon_V \approx \frac{4}{3}e^{-2\sqrt{2/3}\phi/M_{\rm Pl}}$ and $\eta_V \approx -\frac{4}{3}e^{-\sqrt{2/3}\phi/M_{\rm Pl}}$.

b). Show that $N \approx \frac{3}{4}e^{\sqrt{2/3}\phi/M_{\rm Pl}}$, and hence $n_s \approx 1 - \frac{2}{N}$ and $r \approx \frac{12}{N^2}$. Evaluate for $N = 55$.

3. Inflation Solves The Monopole Problem

Many grand unified theories predict that magnetic monopoles of mass $m_M \sim 10^{16}$ GeV were produced around $T \sim 10^{16}$ GeV, with a number density of about one per Hubble volume, $n_M \sim H^3$.

a). Without inflation, estimate n_M/s at production and Ω_M today. Why is this a disaster?

b). Show that N e-folds of inflation after the monopoles are produced reduce their abundance by e^{-3N}. How many e-folds are needed to make it less than one monopole in the observable universe?

4. Eternal De Sitter?

Show that a universe with only a cosmological constant has $\epsilon = 0$ exactly and never stops inflating. Then explain, using the comoving Hubble radius, why an exactly constant H cannot produce the hot Big Bang, and why the small deviation $n_s \neq 1$ is evidence that inflation ended.